PS5: Shared Tasks

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IDAI 610 Fundamentals of AI

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Two annotated corpora for supervised ML

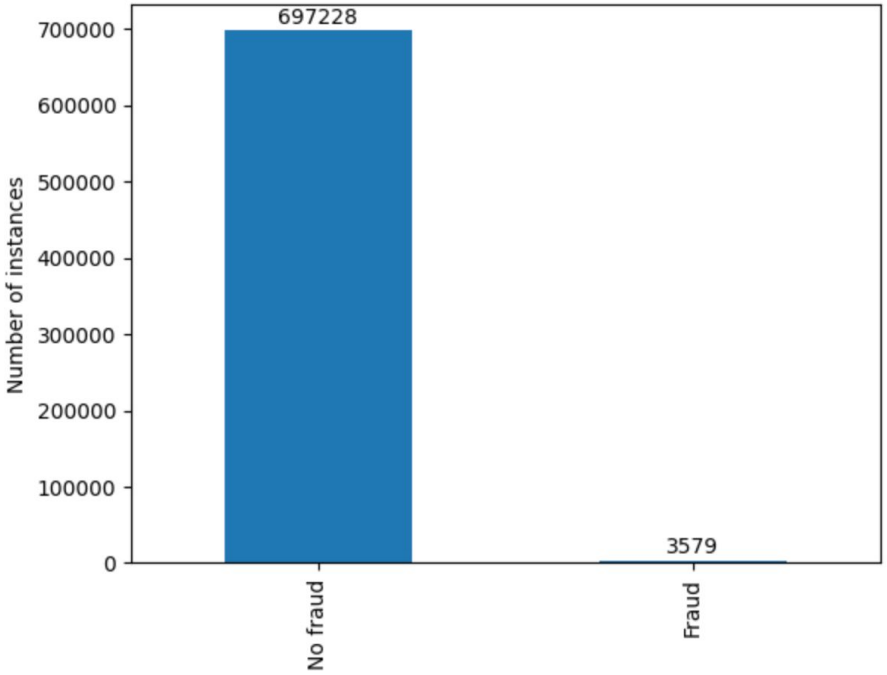
Your team will receive training data from **two** corpora to solve two tasks

- Task 1 uses a credit card transactions to detect fraud transactions
- Task 2 uses a range of sensor data to recognize various human activities

You should not look for either corpus - it would be considered cheating.

Task 1: Credit card fraud

- Synthetically generated dataset of credit card transactions and their corresponding labels
- There are 700807 instances of two different classes (**fraud** and **no fraud**).
- The corpus is *extremely* imbalanced.
- Generated using Sparkov Synthetic Data Generation (i.e., not real data).



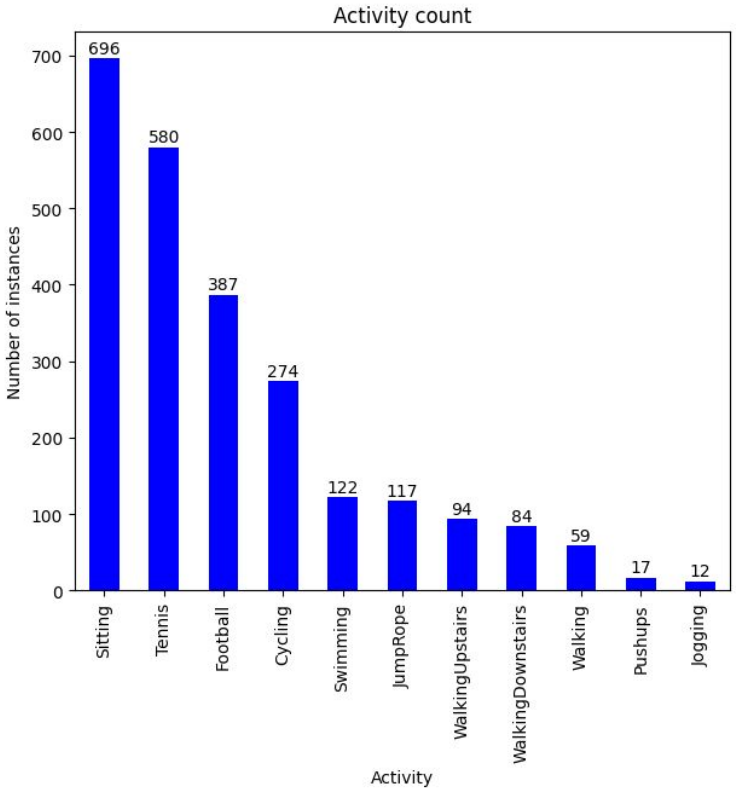
Training set label distribution.

Synthetic credit card fraud dataset: 26 columns

1. **ssn**
2. **cc_num:** credit card number
3. **first**: first name
4. **last**: last name
5. **gender**
6. **street**
7. **city**
8. **state**
9. **zip**
10. **lat**: latitude
11. **long**: longitude
12. **city_pop**: population in the city
13. **job**
14. **dob**: date of birth
15. **acct_num**: account number
16. **profile:** user
17. **trans_num:** transaction number on the credit card
18. **trans_date:** date of transaction
19. **trans_time:** time of transaction
20. **unix_time:** Unix timestamp
21. **category:** purchase category
22. **amt:** amount of purchase
23. **merchant**
24. **merch_lat:** merchant's latitude
25. **merch_long:** merchant's longitude
26. **is_fraud:** *target class*

Task 2: Human activity recognition

- Sensing data (accelerometer, gyroscope, etc.) corresponding to 11 human movement activities (e.g., cycling, walking
- Features are anonymized (will be revealed after submission)
- 2442 instances in the training set
- Imbalanced classes



Training and test data use consistent formats

- Tabular CSV data
- Features:
  - First 2441 columns
  - Features extracted from certain time frames

- Target class labels
  - Last column
  - Label corresp to the time frames

8	9	...	319	320	321	322	323	324	325	326	327	328
35.196034	3.268344	...	0.333333	0.047619	0.095238	0.0	0.0	0.380952	0.333333	0.285714	0.238095	Tennis
143.137411	7.839429	...	0.333333	0.047619	0.095238	0.0	0.0	0.380952	0.333333	0.285714	0.238095	Sitting
143.205373	59.754427	...	0.333333	0.047619	0.095238	0.0	0.0	0.380952	0.333333	0.285714	0.238095	Cycling
0.281777	0.445135	...	0.333333	0.047619	0.095238	0.0	0.0	0.380952	0.333333	0.285714	0.238095	Tennis
44.912627	10.519441	...	0.333333	0.047619	0.095238	0.0	0.0	0.380952	0.333333	0.285714	0.238095	Football

Task 1: Identify the fraudulent transaction

- You will provide a model for a binary classification problem. The model should differentiate between 2 labels (fraud or not). In stage 1, you get access only to training data. You should freeze your final trained system and submit it in deployment mode. You may use any packages you wish. Note: We'll not retrain.

- ***Objective:*** Identify whether a transaction is **fraudulent**, or **not**
  - The gold standard label is in the last column in the training and test data.
  - Training and test sets share the same format.

- Neatly report these metrics on the test set, released in stage 2:
  - Accuracy where **fraud** is the positive class, **not fraud** the negative class (similar to a spam filter)
  - Macro-averaged overall F1 (i.e., both classes are equally weighted)
  - F1 score for the **fraud** class only (do not consider the not fraudulent class)

Task 2: Recognize human activity from provided

- You will provide a model for a multi-class classification problem.
- The model should differentiate between 11 human activities.
- ***Objective***: Identify the type of human activity performed, for a given instance
  - The gold standard label is in the last column in the training and test data.
- Report **accuracy** and **macro-averaged F1** on the later released test set
  - See section 4.7.1 in Jurafsky & Martin, 3rd edition, on how to macro-average metrics

Instructions - read very carefully and ask Qs as needed

1. **The team's notebook should have a standalone, separate prediction python file that calls exported models with a test set and calculates the metrics, i.e., it prints requested metrics for <u>both</u> tasks.**
2. By **Monday, December 1, at 11:59 PM**, submit your team's system implementation.
3. Freeze your code and submit **<u>one</u>** compressed file for the team with:
   - Your system, <u>implemented in **python 3 Jupyter notebook**</u> and the <u>exported models</u> for T1-T2 (so retraining is not required), along with a separate python file with functions for testing and metrics output and <u>instructions</u> on how to import/run the model with test data.
   - A README file with detailed instruction on how to run your system.
   - A `requirements.txt` file only if a package not in **idai610** conda environment
   - No system tuning for test data is allowed. System must be already tuned and stand alone.
4. Test data is released on **Tuesday, December 2, at 7 PM**.
   - For each task, submit your system's output file for test data - for T1 and T2 it should be in the same format as the training and test data but include your predictions instead of gold standard labels.
   - Submit report + output file.
5. You will report test results on your system by **Friday, December 5, at 11:59 PM**.
6. **IMPORTANT: There is no grace period for either phase for this assignment.**
7. We will also run your system with the test data and compare your reported results against our results with your system.

Instructions - continued

6. Please use a Python 3 Jupyter notebook for any code you write in the team.
7. **Use seed numbers in the notebook. We should be able to <u>reproduce</u> your results on the `narnia.gccis.rit.edu` server.**
8. You may use github or another way for sharing code in the team.
9. Do <u>not</u> pass on/distribute data or upload data in a repository where others can access it.
10. If you use other libraries or resources not in `requirements.txt`, we will not perform any installation.
11. Any kind of feature processing or use of external libraries and resources is allowed.
12. Dividing a feature to multiple or combining is allowed and encouraged.
13. You are encouraged to use creative thinking. Data augmentation and manipulation in the training set is allowed.
14. You may use one or more prediction methods or classifiers. For example, multiple classifiers could offer predictions with disagreements resolved by letting classifiers vote to achieve a single joint prediction. You could explore impact on data size.

Group PowerPoint report

Your team should submit a **single** team report **in the form of a PowerPoint** that consists of the following concise sections:

1. Title - including name/acronym of your team's system implementation and the names of contributing team members on the first slide.

2. METHOD - An overview of the team's implementation approach (**What you did**); including both a ***visualization overview*** of your implementation and a ***written description*** of it.

3. RESULTS - performance **per task**:

3.1 RESULTS - **T1: Fraud detection**

A. What is the *baseline*? How did you determine it?
B. A confusion matrix with frequencies (counts), not percentages.
C. Table with accuracy and macro-average F1-score on test data.
D. F1-score for the **fraud** class

3.2 RESULTS - **T2: Human activity recognition**

A. What is the *baseline*?
B. A confusion matrix.
C. Table with accuracy and macro-average F1-score on the test data.

Please submit the team report PowerPoint in the Dropbox by **11:59 pm in the myCourses Dropbox on December 5** also including **your system's output file for test data. Just submit once.**

Note: You already submitted your system and test code on **Monday Dec 1** at time of the code freeze. Do not resubmit.

Group PowerPoint report

Your team should submit a **single** team report **in the form of a PowerPoint** that consists of these concise sections:

5. OBSERVATIONS AND DISCUSSION - of results with interpretation of findings
Also discuss highlights; these can be particularly interesting observations—such as further data analytics, successful specifics of your implementation, especially interesting results, ‘case study’ of particular instances, etc. Include at least one visualization figure.

6. CHALLENGES - issues your team came across and how the team solved them

7. TASK DISTRIBUTION - who did what per task and estimated % effort per person

8. CONCLUSION, followed by REFERENCES as applicable

START IMMEDIATELY AND PRIORITIZE THIS SHORT ASSIGNMENT.

Ideally, you should meet in the group already today or max tomorrow.